Lab 3 - Local search

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Description of the problem

Each instance contains nodes with coordinates (x, y) and a cost. Select exactly half of the nodes (rounded up) and connect them into a closed cycle. Minimise **cycle length + sum of the costs of the selected nodes**. Distances are Euclidean, rounded to the nearest integer.

Algorithms

Eight local search methods, 200 runs each: steepest or greedy × nodes or edges exchange inside the tour × random start (run i uses seed 42 + i) or greedy start (lab 2's best heuristic, greedy cycle with 2-regret + best change, run i starts from node i). Instead of building a tour, they take a complete one and keep improving it with small moves.

Intuition

A construction heuristic never takes a decision back: edges may end up crossing, or a cheap unused node could replace an expensive one. Local search takes a complete tour, applies a small change that lowers the objective, and repeats until no change helps (a local optimum). Example: tour A → C → B → D on a 10 × 10 square has two crossing diagonals (length 48); replacing edges (A, C), (B, D) with (A, B), (C, D) gives Δ = 10 + 10 − 14 − 14 = −8, a tour of 40 that construction would never revisit. **Steepest** applies the best move of the whole neighborhood, **greedy** the first improving one found in random order (more steps, each much cheaper).

Core calculations

Δ = change of the objective, from the changed edges and costs only; p(x), s(x) = predecessor, successor of x.

Inter-route, selected u ↔ unselected v:

```
Δ = c(v) − c(u) + d(p,v) + d(v,s) − d(p,u) − d(u,s)
```

Nodes exchange a ↔ b (for neighbours only the 2 outer edges change):

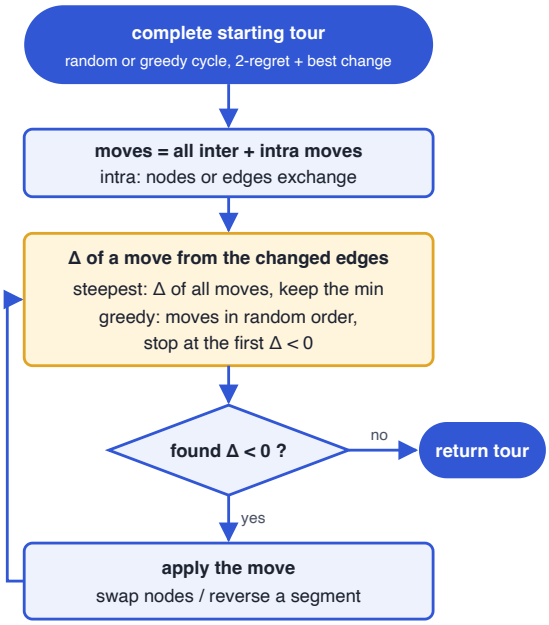
```
Δ = d(pa,b) + d(b,sa) + d(pb,a) + d(a,sb)
− d(pa,a) − d(a,sa) − d(pb,b) − d(b,sb)
```

Edges exchange (a,b), (c,d) → (a,c), (b,d), path b … c reversed:

```
Δ = d(a,c) + d(b,d) − d(a,b) − d(c,d)
```

A move is applied only if Δ < 0; 14 950 moves per step (n = 200, k = 100).

Method	picks	intra-route
Steepest	min Δ of all	nodes / edges
Greedy	first Δ < 0	nodes / edges



Pseudocode

Simplified from `python/solver.py`. All eight methods run `local_search`; they differ in the starting tour (`random_start` / `greedy_start`), in `intra_kind` (`SWAP_NODES` / `SWAP_EDGES`) and in the function that picks the move.

```
def local_search(tour, unselected, search, intra_kind):
    moves = all_moves(k, n - k, intra_kind)    # inter-route + intra-route moves
    while True:
        if search == STEEPEST:
            move = find_best_move(tour, unselected, moves)
        else:
            move = find_first_improving_move(tour, unselected, moves, rng)
        if move is None:
            return tour                         # local optimum
        apply_move(tour, unselected, move)

def find_best_move(tour, unselected, moves):
    best_move, best_delta = None, 0
    for move in moves:
        delta = move_delta(tour, unselected, move)
        if delta < best_delta:
            best_move, best_delta = move, delta
    return best_move

def find_first_improving_move(tour, unselected, moves, rng):
    for tried in range(len(moves)):
        pick = tried + rng.below(len(moves) - tried)
        moves[tried], moves[pick] = moves[pick], moves[tried]
        if move_delta(tour, unselected, moves[tried]) < 0:
            return moves[tried]
    return None

def apply_move(tour, unselected, move):
    kind, i, j = move
    if kind == EXCHANGE_SELECTED: swap tour[i] and unselected[j]
    if kind == SWAP_NODES:        swap tour[i] and tour[j]
    if kind == SWAP_EDGES:        reverse tour[i+1 .. j]
```

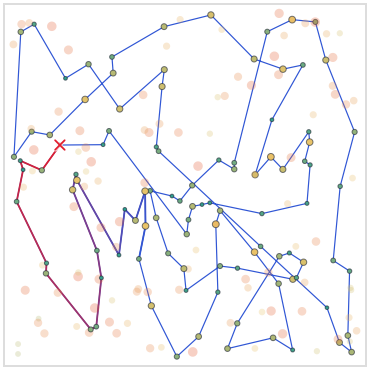
Randomization of the greedy version. A move is a triple (kind, i, j) of positions, and all moves of both kinds are kept in a single list. Each step browses that list with a lazy Fisher-Yates shuffle: the next move is drawn uniformly from the moves not yet tried in this step, and only that move is evaluated. So every order of the whole neighborhood is equally likely, inter- and intra-route moves are fully mixed, and the step stops at the first improving move, so Δ is never computed for the moves it has not reached.

Results

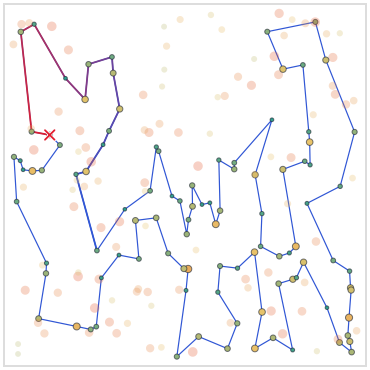
Objective = cycle length + sum of node costs, over 200 runs per method. Each cell is av (min – max); the best value of each kind per instance is in bold.
All best solutions were checked with the solution checker (the same computation as the lecturer's Solution checker.xlsx): **yes**.
Results are identical in all implemented languages: **yes**.

Method	TSPA	TSPB
Steepest LS, nodes exchange, random start	89742 (81937 – 96386)	66055 (59059 – 74933)
Steepest LS, edges exchange, random start	76225 (74248 – 79388)	**52461** (50265 – **55173**)
Greedy LS, nodes exchange, random start	87889 (81686 – 95320)	63656 (56270 – 70172)
Greedy LS, edges exchange, random start	76265 (73545 – 78428)	52913 (**49955** – 55677)
Steepest LS, nodes exchange, greedy start	73018 (**71955** – **74682**)	57376 (52778 – 63043)
Steepest LS, edges exchange, greedy start	72997 (**71955** – **74682**)	57019 (52566 – 62301)
Greedy LS, nodes exchange, greedy start	73015 (**71955** – **74682**)	57379 (52672 – 63043)
Greedy LS, edges exchange, greedy start	**72971** (**71955** – **74682**)	57067 (52672 – 62500)

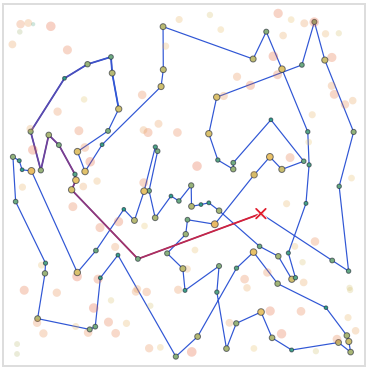
Visualisation (Instance TSPA)



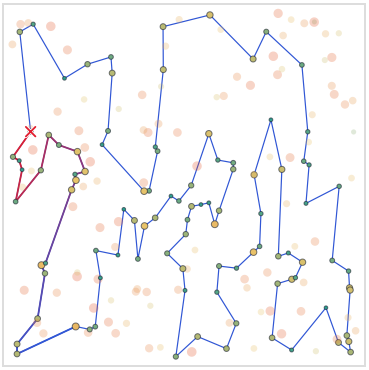
Steepest LS, nodes exchange, random start
81937 (length 30984 + cost 50953)



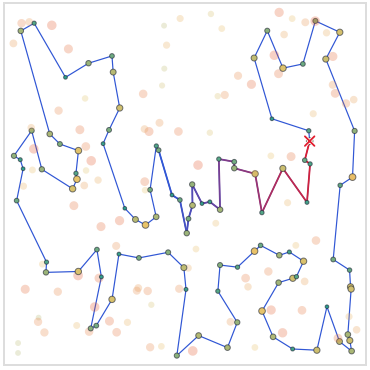
Steepest LS, edges exchange, random start
74248 (length 22259 + cost 51989)



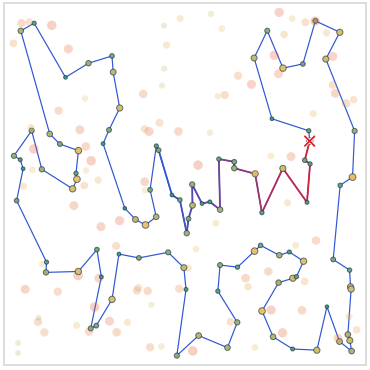
Greedy LS, nodes exchange, random start
81686 (length 30012 + cost 51674)



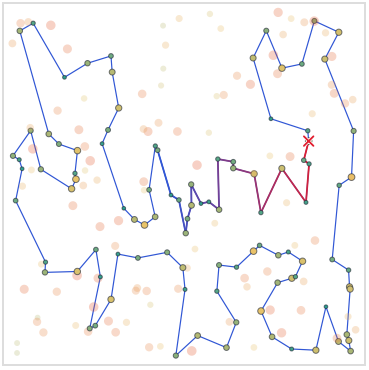
Greedy LS, edges exchange, random start
73545 (length 22205 + cost 51340)



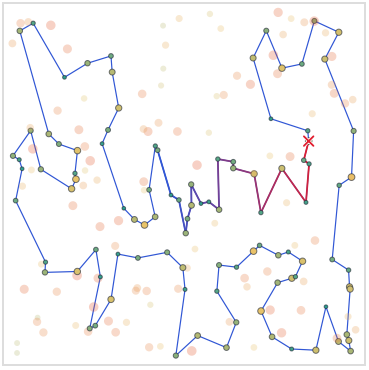
Steepest LS, nodes exchange, greedy start
71955 (length 21123 + cost 50832)



Steepest LS, edges exchange, greedy start
71955 (length 21123 + cost 50832)

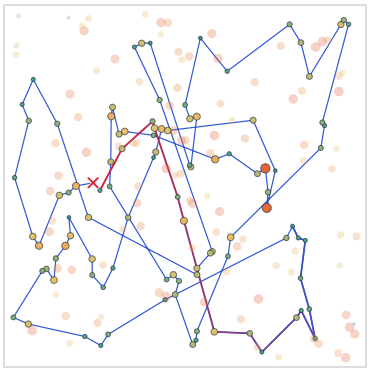


Greedy LS, nodes exchange, greedy start
71955 (length 21123 + cost 50832)

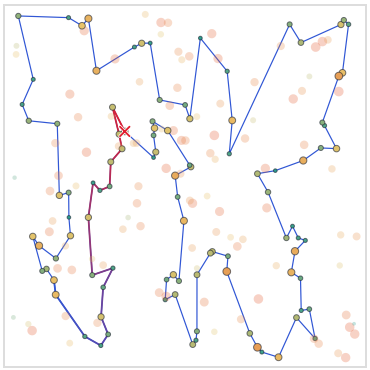


Greedy LS, edges exchange, greedy start
71955 (length 21123 + cost 50832)

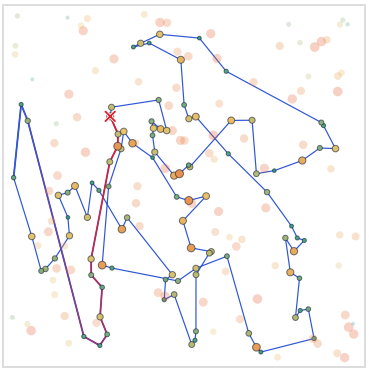
Visualisation (Instance TSPB)



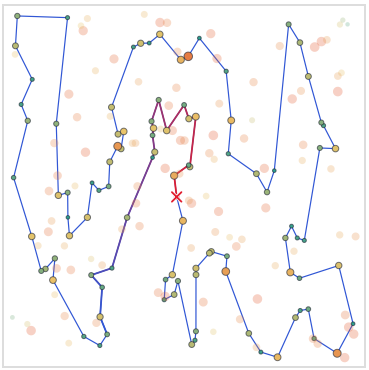
Steepest LS, nodes exchange, random start
59059 (length 28949 + cost 30110)



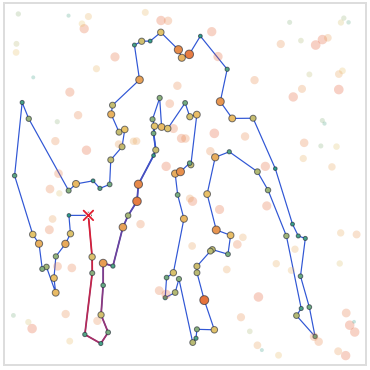
Steepest LS, edges exchange, random start
50265 (length 19741 + cost 30524)



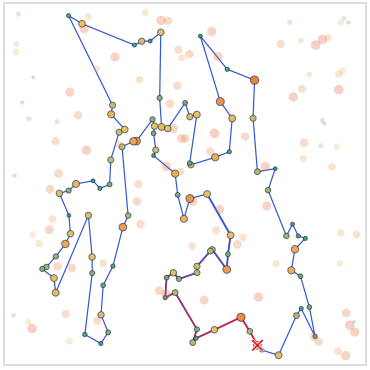
Greedy LS, nodes exchange, random start
56270 (length 23242 + cost 33028)



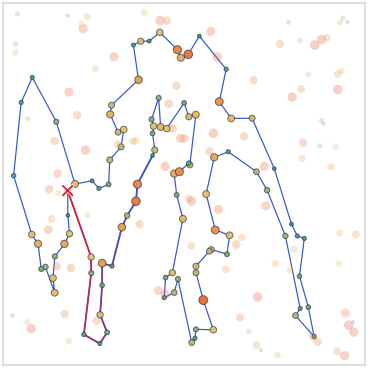
Greedy LS, edges exchange, random start
49955 (length 20215 + cost 29740)



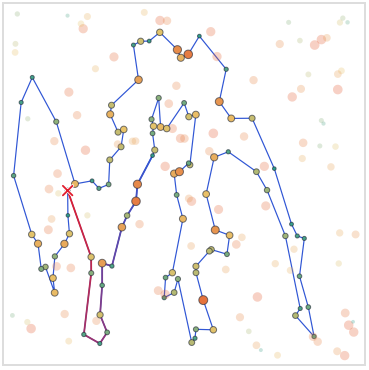
Steepest LS, nodes exchange, greedy start
52778 (length 16865 + cost 35913)



Steepest LS, edges exchange, greedy start
52566 (length 17522 + cost 35044)



Greedy LS, nodes exchange, greedy start
52672 (length 17168 + cost 35504)



Greedy LS, edges exchange, greedy start
52672 (length 17168 + cost 35504)

Running time

Average time of one run of the algorithm only, over the same runs as above. The fastest language in each row is in bold.

Instance TSPA

Method	Python	C	C++	Rust
Steepest LS, nodes exchange, random start	336.9 ms	4.69 ms	**4.31 ms**	6.62 ms
Steepest LS, edges exchange, random start	255.2 ms	3.48 ms	**3.38 ms**	4.13 ms
Greedy LS, nodes exchange, random start	70.59 ms	1.29 ms	**1.26 ms**	1.37 ms
Greedy LS, edges exchange, random start	44.41 ms	0.8554 ms	**0.8143 ms**	0.8704 ms
Steepest LS, nodes exchange, greedy start	45.03 ms	0.9084 ms	**0.8863 ms**	0.9632 ms
Steepest LS, edges exchange, greedy start	43.54 ms	0.8906 ms	**0.8749 ms**	0.9478 ms
Greedy LS, nodes exchange, greedy start	52.27 ms	1.08 ms	**1.05 ms**	1.13 ms
Greedy LS, edges exchange, greedy start	50.53 ms	1.06 ms	**1.04 ms**	1.11 ms

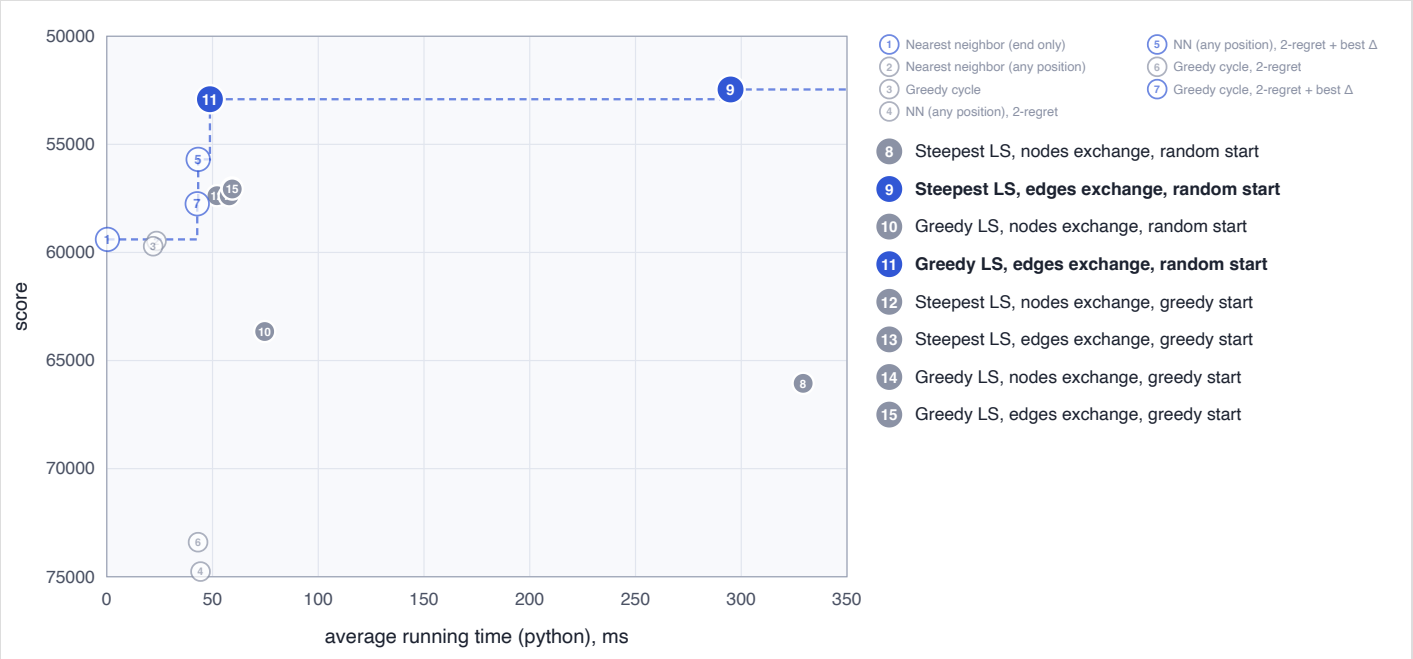
Pareto plot (Instance TSPA)



Instance TSPB

Method	Python	C	C++	Rust
Steepest LS, nodes exchange, random start	329.3 ms	4.57 ms	**4.37 ms**	5.46 ms
Steepest LS, edges exchange, random start	294.9 ms	3.54 ms	**3.44 ms**	4.28 ms
Greedy LS, nodes exchange, random start	74.65 ms	1.34 ms	**1.27 ms**	1.29 ms
Greedy LS, edges exchange, random start	48.77 ms	**0.8300 ms**	0.8732 ms	0.9377 ms
Steepest LS, nodes exchange, greedy start	52.11 ms	1.05 ms	**1.00 ms**	1.10 ms
Steepest LS, edges exchange, greedy start	59.17 ms	1.12 ms	**1.04 ms**	1.16 ms
Greedy LS, nodes exchange, greedy start	57.89 ms	1.17 ms	**1.12 ms**	1.23 ms
Greedy LS, edges exchange, greedy start	59.31 ms	1.21 ms	**1.18 ms**	1.26 ms

Pareto plot (Instance TSPB)



Best solutions (0-based node indices)

Instance TSPA

Steepest LS, nodes exchange, random start
152, 49, 149, 63, 168, 97, 127, 171, 2, 187, 30, 160, 123, 83, 141, 180, 28, 156, 102, 87, 98, 114, 181, 58, 99, 131, 72, 151, 57, 182, 192, 158, 150, 106, 113, 89, 42, 134, 84, 135, 73, 45, 185, 35, 67, 144, 52, 33, 54, 9, 136, 163, 29, 101, 147, 47, 130, 159, 153, 78, 88, 107, 176, 16, 51, 139, 112, 4, 91, 80, 48, 31, 116, 60, 108, 38, 71, 157, 115, 39, 186, 184, 178, 164, 8, 142, 85, 1, 169, 3, 50, 93, 174, 36, 177, 179, 53, 140, 76, 132

Steepest LS, edges exchange, random start
108, 60, 31, 48, 80, 38, 91, 157, 71, 4, 76, 132, 90, 123, 30, 180, 35, 51, 16, 67, 144, 140, 53, 179, 52, 177, 36, 99, 131, 33, 54, 9, 164, 8, 174, 176, 150, 106, 43, 85, 3, 50, 169, 1, 178, 184, 136, 29, 101, 147, 47, 93, 130, 159, 146, 170, 75, 153, 193, 78, 88, 107, 113, 89, 42, 151, 57, 182, 111, 138, 72, 134, 84, 58, 158, 192, 181, 114, 135, 61, 73, 45, 185, 28, 87, 141, 187, 2, 171, 92, 69, 127, 97, 168, 116, 149, 63, 11, 49, 152

Greedy LS, nodes exchange, random start
174, 87, 160, 83, 123, 152, 108, 49, 60, 80, 91, 157, 71, 4, 76, 100, 90, 132, 139, 112, 115, 186, 136, 184, 1, 50, 93, 106, 89, 42, 150, 176, 131, 36, 177, 179, 52, 144, 67, 185, 35, 16, 51, 156, 28, 180, 30, 161, 11, 63, 149, 116, 168, 97, 127, 69, 171, 2, 187, 141, 114, 181, 134, 72, 151, 107, 153, 193, 78, 88, 57, 182, 138, 158, 192, 58, 84, 135, 73, 45, 140, 53, 99, 8, 142, 85, 3, 164, 9, 54, 33, 143, 62, 178, 29, 101, 147, 47, 159, 130

Greedy LS, edges exchange, random start
60, 116, 149, 63, 168, 49, 108, 152, 100, 90, 123, 83, 160, 97, 95, 127, 69, 129, 104, 92, 171, 2, 187, 30, 141, 180, 28, 87, 102, 185, 67, 144, 52, 143, 33, 9, 54, 131, 99, 36, 177, 179, 53, 140, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 174, 8, 164, 85, 150, 106, 113, 89, 42, 151, 182, 57, 107, 88, 78, 193, 153, 170, 146, 159, 130, 47, 93, 50, 3, 1, 178, 136, 186, 39, 115, 112, 51, 16, 35, 156, 132, 76, 71, 157, 91, 80, 48, 31

Steepest LS, nodes exchange, greedy start
169, 3, 50, 93, 85, 174, 8, 54, 9, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48, 31, 108, 152, 100, 123, 83, 160, 49, 60, 116, 149, 63, 168, 97, 127, 161, 30, 187, 171, 2, 189, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 170, 146, 159, 130, 47, 37, 147, 101, 55, 29, 178, 184, 136, 186, 164, 1

Steepest LS, edges exchange, greedy start
169, 3, 50, 93, 85, 174, 8, 54, 9, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48, 31, 108, 152, 100, 123, 83, 160, 49, 60, 116, 149, 63, 168, 97, 127, 161, 30, 187, 171, 2, 189, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 170, 146, 159, 130, 47, 37, 147, 101, 55, 29, 178, 184, 136, 186, 164, 1

Greedy LS, nodes exchange, greedy start
169, 3, 50, 93, 85, 174, 8, 54, 9, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48, 31, 108, 152, 100, 123, 83, 160, 49, 60, 116, 149, 63, 168, 97, 127, 161, 30, 187, 171, 2, 189, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 170, 146, 159, 130, 47, 37, 147, 101, 55, 29, 178, 184, 136, 186, 164, 1

Greedy LS, edges exchange, greedy start
169, 3, 50, 93, 85, 174, 8, 54, 9, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48, 31, 108, 152, 100, 123, 83, 160, 49, 60, 116, 149, 63, 168, 97, 127, 161, 30, 187, 171, 2, 189, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 170, 146, 159, 130, 47, 37, 147, 101, 55, 29, 178, 184, 136, 186, 164, 1

Instance TSPB

Steepest LS, nodes exchange, random start

154, 140, 21, 133, 136, 151, 98, 88, 99, 171, 74, 20, 2, 14, 153, 90, 129, 127, 80, 85, 45, 134, 175, 174, 1, 132, 29, 197, 6, 180, 115, 170, 155, 71, 60, 40, 64, 69, 160, 8, 117, 104, 54, 106, 192, 159, 162, 76, 135, 172, 183, 30, 130, 191, 38, 47, 59, 73, 167, 51, 82, 87, 179, 67, 66, 10, 114, 196, 105, 94, 81, 187, 70, 190, 57, 89, 72, 46, 86, 3, 184, 96, 198, 121, 120, 24, 34, 152, 181, 97, 158, 149, 28, 145, 65, 178, 103, 177, 125, 26

Steepest LS, edges exchange, random start

183, 130, 30, 21, 38, 47, 140, 154, 97, 71, 40, 60, 157, 45, 134, 175, 36, 197, 29, 132, 178, 103, 6, 115, 170, 125, 177, 158, 145, 28, 149, 37, 116, 147, 42, 50, 121, 198, 96, 120, 72, 46, 89, 57, 185, 76, 190, 70, 81, 94, 105, 61, 139, 196, 114, 148, 10, 78, 54, 162, 192, 80, 127, 129, 90, 52, 182, 153, 20, 14, 2, 74, 17, 171, 39, 99, 31, 67, 24, 34, 181, 179, 87, 82, 51, 85, 59, 73, 167, 98, 151, 122, 3, 184, 117, 133, 136, 172, 160, 69

Greedy LS, nodes exchange, random start

191, 30, 176, 38, 155, 71, 60, 157, 45, 134, 175, 145, 28, 65, 178, 132, 29, 6, 115, 170, 177, 125, 26, 97, 154, 140, 33, 64, 73, 59, 85, 51, 82, 87, 179, 181, 34, 24, 25, 98, 27, 111, 151, 69, 189, 183, 21, 47, 141, 40, 167, 152, 67, 99, 39, 171, 2, 14, 20, 74, 153, 182, 80, 52, 90, 129, 127, 192, 76, 86, 46, 72, 156, 96, 121, 198, 48, 89, 57, 196, 114, 148, 10, 78, 54, 162, 104, 185, 3, 184, 173, 122, 160, 172, 136, 133, 8, 117, 120, 130

Greedy LS, edges exchange, random start

151, 122, 184, 3, 86, 46, 72, 117, 120, 133, 136, 172, 160, 69, 64, 40, 71, 60, 157, 45, 134, 175, 197, 6, 29, 132, 178, 65, 145, 28, 149, 146, 37, 116, 158, 177, 125, 170, 115, 97, 154, 140, 47, 38, 176, 21, 183, 30, 130, 121, 198, 96, 48, 156, 32, 89, 57, 185, 76, 162, 192, 54, 190, 70, 187, 196, 114, 148, 10, 90, 129, 127, 80, 182, 153, 118, 110, 93, 2, 14, 20, 74, 17, 171, 99, 31, 67, 24, 34, 152, 181, 179, 87, 82, 167, 51, 85, 59, 73, 98

Steepest LS, nodes exchange, greedy start

97, 155, 71, 175, 134, 45, 157, 60, 141, 40, 33, 64, 4, 124, 69, 160, 172, 136, 133, 120, 8, 117, 72, 46, 86, 3, 184, 173, 122, 151, 98, 73, 59, 85, 51, 167, 82, 87, 179, 88, 126, 181, 152, 34, 24, 67, 66, 11, 27, 135, 76, 162, 192, 80, 20, 74, 2, 14, 153, 90, 129, 127, 54, 104, 185, 79, 57, 89, 32, 156, 43, 48, 96, 198, 121, 63, 130, 191, 30, 183, 21, 38, 47, 140, 154, 26, 125, 145, 28, 65, 178, 103, 132, 29, 36, 197, 6, 180, 115, 170

Steepest LS, edges exchange, greedy start

39, 99, 163, 88, 87, 82, 179, 51, 85, 59, 73, 167, 181, 152, 34, 24, 31, 67, 66, 27, 111, 98, 151, 122, 69, 160, 172, 136, 133, 168, 189, 21, 64, 33, 40, 60, 157, 45, 134, 175, 71, 155, 97, 36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 154, 140, 47, 38, 30, 183, 191, 130, 116, 147, 121, 198, 96, 48, 120, 8, 117, 72, 46, 86, 184, 3, 135, 76, 185, 79, 89, 57, 77, 104, 162, 54, 192, 80, 127, 129, 90, 52, 182, 153, 14, 2, 20, 74, 17, 171

Greedy LS, nodes exchange, greedy start

125, 155, 71, 175, 134, 45, 157, 60, 141, 40, 33, 64, 4, 124, 69, 160, 172, 136, 133, 120, 8, 117, 72, 46, 86, 3, 184, 173, 122, 151, 98, 73, 59, 85, 51, 167, 82, 87, 179, 88, 126, 181, 152, 34, 24, 67, 66, 11, 27, 135, 76, 162, 192, 80, 20, 74, 2, 14, 153, 90, 129, 127, 54, 104, 185, 79, 57, 89, 32, 156, 43, 48, 96, 198, 121, 63, 130, 191, 30, 183, 21, 38, 47, 140, 154, 26, 158, 149, 28, 65, 178, 103, 132, 29, 36, 197, 6, 180, 115, 170

Greedy LS, edges exchange, greedy start

125, 155, 71, 175, 134, 45, 157, 60, 141, 40, 33, 64, 4, 124, 69, 160, 172, 136, 133, 120, 8, 117, 72, 46, 86, 3, 184, 173, 122, 151, 98, 73, 59, 85, 51, 167, 82, 87, 179, 88, 126, 181, 152, 34, 24, 67, 66, 11, 27, 135, 76, 162, 192, 80, 20, 74, 2, 14, 153, 90, 129, 127, 54, 104, 185, 79, 57, 89, 32, 156, 43, 48, 96, 198, 121, 63, 130, 191, 30, 183, 21, 38, 47, 140, 154, 26, 158, 149, 28, 65, 178, 103, 132, 29, 36, 197, 6, 180, 115, 170

Source code

labs/lab3 (Python, C, C++, Rust)

Conclusions

- Edges exchange is much better than nodes exchange from random starts (average TSPA 76 225 vs 89 742, TSPB 52 461 vs 66 055), because one 2-opt move can remove a crossing; from greedy starts all four variants end within 1 % of each other, since the start is already close to a local optimum.
- Local search beats every earlier method: the best average drops from 73 350 to 72 971 on TSPA (greedy LS, edges, greedy start) and from 55 698 to 52 461 on TSPB (steepest LS, edges, random start), where random starts with edges exchange beat the greedy starts.
- Greedy LS reaches the same quality as steepest LS about 4× faster from random starts (C: 0.86 ms vs 3.48 ms on TSPA, edges), and local search after the greedy start adds almost nothing to the ~1 ms of the construction.